Adapting Segment Anything Model for Change Detection in VHR Remote Sensing Images

Lei Ding, Kun Zhu, Daifeng Peng, Hao Tang, Kuiwu Yang, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*— Vision foundation models (VFMs), such as the segment anything model (SAM), allow zero-shot or interactive segmentation of visual contents; thus, they are quickly applied in a variety of visual scenes. However, their direct use in many remote sensing (RS) applications is often unsatisfactory due to the special imaging properties of RS images (RSIs). In this work, we aim to utilize the strong visual recognition capabilities of VFMs to improve change detection (CD) in very high-resolution (VHR) RSIs. We employ the visual encoder of FastSAM, a variant of the SAM, to extract visual representations in RS scenes. To adapt FastSAM to focus on some specific ground objects in RS scenes, we propose a convolutional adaptor to aggregate the task-oriented change information. Moreover, to utilize the semantic representations that are inherent to SAM features, we introduce a task-agnostic semantic learning branch to model the semantic latent in bitemporal RSIs. The resulting method, SAM-based CD (SAM-CD), obtains superior accuracy compared with the state-of-the-art (SOTA) fully supervised CD methods and exhibits a sample-efficient learning ability that is comparable to semisupervised CD methods. To the best of our knowledge, this is the first work that adapts VFMs to CD in VHR RSIs.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), remote sensing (RS), segment anything model (SAM), vision foundation models (VFMs).**

I. Introduction

RECENTLY, vision foundation models (VFMs) have emerged and gained great research focus in the field of computer vision. Utilizing the knowledge gained in meta-scale datasets, foundation models, such as segment anything model (SAM) [1] and its variants (e.g., FastSAM [2] and Mobile SAM [3]) are able to recognize visual contents in a training-free manner and obtain fine-grained semantic masks. Their strong generalization across different imaging conditions and visual objects promotes greatly the applications in real-world scenarios. However, due to the inductive bias learned in natural images, foundational models exhibit limitations when applied to images in some specific domains, such as medical images and remote sensing images (RSIs). According to the literature survey [4], SAM pays more attention to the foreground objects and often fails to segment small and irregular objects. In this article, we focus on adapting SAM to improve one of the fundamental tasks in RS, i.e., change detection (CD) of very high-resolution (VHR) RSIs.

CD is the task of segmenting content changes in multitemporal RSIs. It is crucial for a wide range of real-world applications, including environmental monitoring, urban management, disaster alerting, land cover/land use (LCLU) monitoring, and so on. With the continuous development of Earth observation technologies and deep learning methods, we are now able to monitor Earth's surface with large volumes of HR RSIs and automatically detect the changes. In particular, deep neural networks, including convolutional neural networks (CNNs) and vision transformers (ViTs) [5], are widely used for CD in VHR RSIs. A common practice is to use weight-sharing CNNs, i.e., Siamese CNNs, to aggregate and segment the multitemporal semantic changes. ViTs are also leveraged as encoders to aggregate the features or to model the semantic-change correlations. State-of-the-art (SOTA) methods can detect changes with high accuracy.

Although much progress has been made, the use of CD techniques in real-world applications is still limited. The deep learning-based methods generally require large volumes of high-quality training data, whereas it is difficult to collect enough well-annotated change instances. The construction of a large CD training set requires that there is a certain time gap between the acquisition dates, and the observation platform covers large areas. For some small or rare LCLU classes, it is often difficult to collect enough training samples. Moreover, in multitemporal VHR RSIs, there are often differences in imaging conditions, such as illumination conditions, imaging angle, observation seasons, or even sensing platforms that are specific to each dataset. Therefore, given only change samples but no semantic information, it is still challenging to discriminate the semantic changes from temporal differences.

In this article, we aim to utilize the strong and universal semantic exploitation capability of SAM to boost CD accuracy and reduce the dependence on large volumes of training samples. The major contributions can be summarized as follows.

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Lei Ding and Kuiwu Yang are with the Department of Big Data Analysis, PLA Strategic Support Force Information Engineering University, Zhengzhou 450001, China (e-mail: dinglei14@outlook.com; yangkw@aliyun.com).

Kun Zhu is with the Institute of Geospatial Information, PLA Strategic Support Force Information Engineering University, Zhengzhou 450001, China (e-mail: zkun@whu.edu.cn).

Daifeng Peng is with the School of Remote Sensing and Geomatics Engineering, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: daifeng@nuist.edu.cn).

Hao Tang is with the Department of Information Technology and Electrical Engineering, ETH Zürich, 8092 Zürich, Switzerland (e-mail: hao.tang@vision.ee.ethz.ch).

Lorenzo Bruzzone is with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: lorenzo.bruzzone@unitn.it).

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1) Introducing VFMs to the CD of VHR RSIs and proposing a SAM-based CD (SAM-CD) network. The SAM-CD adapts the FastSAM, an efficient variant of the SAM, to RS scenes and utilizes the pretrained priors to embed multitemporal LCLU representations. To the best of our knowledge, this is the first work that introduces and adapts VFMs to the CD task. Experimental results reveal that SAM-CD obtains significant accuracy improvements over the SOTA methods and exhibits a sample-efficient learning ability that is comparable to semisupervised CD methods.
2) Introducing a task-agnostic learning of semantic latent in CD of VHR RSIs. Different from conventional CD approaches that exploit only difference information, we propose to model CD in a multitask learning perspective and introduce an explicit semantic learning branch in the SAM-CD. Furthermore, a temporal constraint-based learning objective is proposed to drive the network toward embedding consistent semantic representations in multitemporal features. The resulting approach can better discriminate semantic changes and does not require explicit semantic supervision.

The remainder of this article is organized as follows. Section II reviews the literature works related to CD and VFMs. In Section III, we introduce the proposed SAM-CD. Section IV reports the experimental settings and the results obtained on a benchmark dataset. Section V draws conclusions to this study.

II. Related Work

This section first reviews CD methods using deep neural networks and then introduces the development of VFMs.

A. CD in RSIs

Before deep learning gained prominence, CD methods extracted and analyzed the change features to segment changes. They can be categorized into three categories based on the types of analyzed features, including texture features, object-based features, and angular features [6].

In recent years, CNNs have been extensively utilized for CD due to their ability to capture contextual information in RSIs. Initially, deep CNN-based CD was treated as a segmentation task, where UNet-like CNN models are employed to directly segment the changed objects [7]. In [8], a common CNN framework for CD is established. It utilizes two Siamese CNN encoders to extract the multitemporal features, before embedding them into change representations with a CNN decoder.

The major challenges in CD on VHR RSIs are to distinguish semantic changes between seasonal changes and mitigating spatial misalignment as well as illumination differences. In CNN-based methods, channelwise feature difference operations are commonly used to extract the change features [8], [9]. Another common strategy is to leverage multiscale features to reduce the impact of redundant spatial details [10]. Multiscale binary supervisions are also introduced in [7] to align the embedding of change features. Recently, the attention mechanism has been widely used to model the spatial context in RSIs [11]. Channelwise attention is often used to improve the change representations [12], [13], while spatial attention is often used to exploit the long-range context dependencies [14], [15]. Apart from these techniques, change augmentation (i.e., synthesizing change samples) is a commonly used strategy in single-temporal CD to supervise the learning of semantic changes. Typical approaches include generating change instances [16] or removing objects [17] in VHR RSIs.

Another research focus in CD is to model the temporal dependencies in the pairs of RSIs. In [18], hybrid CNN and recurrent neural networks (RNNs) are proposed for multiclass CD in RSIs, where the RNN modules are leveraged to reason the temporal information. In [19], a multilayer RNN module is adopted to learn change probabilities. In [20], a cross-time self-attention is proposed to model the implicit semantic correlations in multitemporal features. Graph convolutional networks are also an efficient technique to propagate LCLU information and, thus, are often adopted in semisupervised [21] and unsupervised CD methods [22].

Recently, ViTs [5] gained great research interests in visual tasks. There are two strategies to utilize ViTs for CD of VHR RSIs. The first one is to replace CNN backbones with ViTs to extract temporal features, such as ChangeFormer [23] and edge-assisted adaptive transformer detector (EATDer) [24]. Meanwhile, ViTs can also be used to model the temporal dependencies. In bitemporal image transformer (BIT) [25], a transformer encoder is employed to extract changes of interest, while two Siamese transformer decoders are placed to refine the change maps. In cross-temporal difference transformer (CTD-Former) [26], a cross-temporal transformer is proposed to interact between the different temporal branches. In [27], a SCanFormer is introduced to model jointly the representation of semantic and change embeddings.

B. Vision Foundation Models

One of the bottleneck problems in applying deep neural networks in real-world applications is their requirement for a large amount of well-annotated training data, especially for dense prediction tasks, such as semantic segmentation and CD. To address this problem, researchers have explored to pretrain vision models with web-scale datasets to obtain universal recognition capability that can be generalized to downstream tasks. A notable example is contrastive language-image pretraining (CLIP) [28], a model that can describe visual content with text descriptions, trained with 400 million image–text pairs. Its zero-shot image classification accuracy is comparable to a fully supervised CNN. In [29], denoising diffusion probabilistic models are pretrained on large-scale RS datasets before they serve as feature extractors in CD of VHR RSIs.

An emerging trend in computer vision is to explore foundation models that can be generalized to common vision tasks with specified user prompts. This is sparked by the SAM [1], a segmentation model that is trained on millions of annotated images to gain zero-shot generalization to "unseen" images and objects. Given user prompts indicating the location or text descriptions, SAM is able to segment the objects with interests during the inference. Similar to SAM, segment everything with a generalist painter (SegGPT) [30] also claims zero-shot recognition capability on common vision images. Furthermore, segment everything everywhere with multi-modal prompts (SEEM) [31] enables more flexible user prompts, including points, scribbles, text, audio, and their combinations. Since

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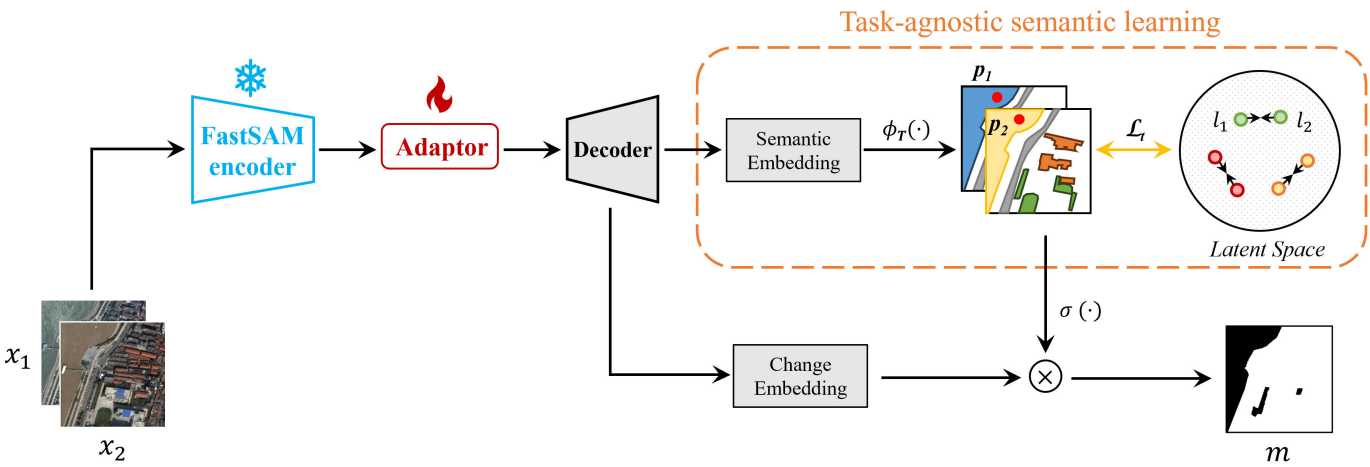


Fig. 1. Architecture of the proposed SAM-CD.

SAM is expensive in computation resources, FastSAM [2] is proposed to segment anything in real time. Its generalization performance is comparable to SAM, while its inference speed is 50 times faster than SAM.

Although VFMs claim to be able to "segment anything" [1], they exhibit significant limitations in certain domains, including medical images, manufacturing scenes, and RSIs [4]. Since VFMs are mostly trained on natural images, they tend to focus more on the foreground objects and struggle to segment small and irregular objects. Another limitation is that VFMs do not provide semantic classes associated with the masks. A feasible way to segment RSIs is to use text prompts to generate segmentation samples and then progressively map all the classes [32]. In this work, we utilize adaption to fine-tune the VFM to learn semantic latent in VHR RSIs.

III. Proposed Method

This section describes the proposed SAM-CD architecture that adapts SAM to CD in VHR RSIs. First, we present an overview of the network. Then, we elaborate on the technical details of the FastSAM adaptor and the task-agnostic semantic learning. Finally, we report the implementation details.

It is worth noting that the foundation model in SAM-CD is FastSAM, since it supports access to low-level spatial features and requires fewer computational resources. However, it is possible to replace FastSAM with other VFMs in the SAM-CD architecture.

A. Overview

Typical deep learning-based methods for CD in RSIs leverage two Siamese encoders to extract temporal features and then embed them into change representations with a shared decoder [7], [8]. Let (x_1, x_2) denote the pair of temporal images, and $U(\cdot)$ and $V(\cdot)$ denote the encoder and decoder networks. We can write

$$m = V[U(x_1, x_2)] \tag{1}$$

where m is a binary CD map that highlights the changes. A limitation of this architecture is that both $U(\cdot)$ and $V(\cdot)$ mainly focus on the temporal differences. Differently, we expect that the semantic changes can be learned better by comparing the underlying semantic features. With VFMs, it is now possible to extract the semantics of ground objects without categorical annotations. In SAM-CD, we extract the semantic latent to better discriminate the categorical changes. In other words, we employ VFM as visual encoders, denoted as $\widetilde{U}(\cdot)$, to extract universal semantic features (instead of temporal differences), while the $V(\cdot)$ compares the extracted features. This can be written as follows:

$$m = V\left[\widetilde{U}(x_1), \widetilde{U}(x_2)\right]. \tag{2}$$

Fig. 1 presents an overview of the proposed SAM-CD architecture. First, we leverage FastSAM as a frozen encoder to exploit the visual entities. For better generalization in RSIs, a trainable adaptor is introduced to adapt the extracted features. This is described in more detail in Section III-B. The obtained multiscale FastSAM features are fused and upsampled in a UNet-like convolutional decoder. Then, apart from the change branch that embeds change representations, we introduce an additional task-agnostic semantic learning branch to model the underlying semantic latent. This is illustrated in Section III-C. The resulting SAM-CD is semantic-aware; thus, it can better capture the object changes in VHR RSIs.

B. FastSAM Adaptor

Although VFMs are capable of extracting semantic representations from any optical image, they exhibit limitations in discriminating ground objects in RSIs [4]. In the field of natural language processing [33], adaption is a commonly used strategy to fine-tune the pretrained models to downstream tasks. Since the FastSAM is constructed with CNNs, we adopt convolutional adaptors to adapt the extracted features.

The inner structure of the proposed adaptor network is illustrated in Fig. 2. First, we aggregate the features extracted by FastSAM at the spatial scales of 1/32, 1/16, 1/8, and 1/4, denoted as f_1, f_2, f_3, and f_4. Each feature f_i is processed by

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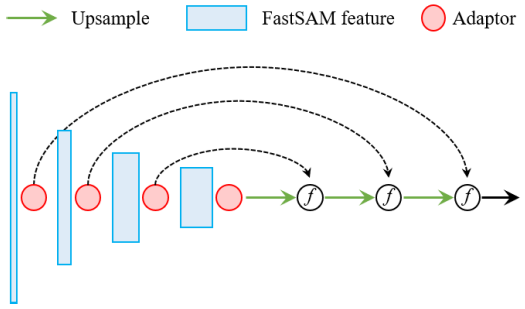


Fig. 2. Proposed adaptor network to utilize FastSAM features. Each (f) denotes a convolutional fusion operation.

a corresponding adaptor α, denoted as follows:

$$f_i^* = \alpha(f_i) = \gamma\{\text{bn}[\text{conv}(f_i)]\} \tag{3}$$

where conv denotes a 1×1 convolutional layer, bn denotes a batch normalization function, and $\gamma()$ is a rectified linear unit (ReLU) function. Since there are fewer object categories in RSIs than those in natural images, we reduce the channel number of f_i^* to reduce redundancy.

In CD of VHR RSIs, low-level features are important for segmenting the areas with changes [7], [8]. Therefore, we employ a UNet-like decoder to fuse the multiscale features after adaptation. For each level of feature f_i, we fuse it with the lower level feature f_{i+1} in a decoder block, denoted as follows:

$$d_1 = f_1, \quad d_{i+1} = \text{conv}[f_{i+1}, \text{upsample}(d_i)] \tag{4}$$

where d_i is the ith layer of feature in the decoder. We then concatenate the resulting features $\{d_1, d_2, d_3\}$ to obtain semantic representations that are adaptive to the RS domain.

C. Task-Agnostic Semantic Learning

The literature reveals that joint learning of multiple related tasks can improve the performance of each single task [34], [35]. To boost the performance of CD, we introduce an additional temporal semantic learning branch. This is similar to the strategy of post-classification CD used in the last decades [36], i.e., segmenting changes after classifying the multitemporal RSIs. However, in the proposed SAM-CD architecture, the learning of semantics and changes is conducted simultaneously.

In Section III-B, we described the adapted SAM features $\{d_1, d_2, d_3\}$. We further use convolutional operations to transform them into a candidate latent $\hat{l} \in \mathbb{R}^{k \times w \times h}$, where k indicates the number of interesting semantic clusters in RSIs. Supervising the learning of this latent with change labels will drive the network to focus on only the change classes, i.e., being task-specific. Instead, we use the underlying temporal constraints to supervise the learning of semantic latent.

Different from the literature research that explicitly supervises the learning of LCLU categories [20], in binary CD task, the semantic labels of each acquisition date are not available. Therefore, the SAM-CD implicitly supervises the learning of bitemporal latent by aligning their feature representations. For each candidate latent $\hat{l}$, a softmax function ϕ is used to normalize them

$$\phi_T(\hat{l}_i) = \frac{e^{\hat{l}_i/T}}{\sum_{j=1}^{n} e^{\hat{l}_j/T}} \tag{5}$$

where T is the temperature parameter to control the probability distribution of output features. We set $T > 1$ to obtain more diverse semantic representations. The selection of the value of T is discussed in Section IV. Let $\{l_1, l_2\}$ be the normalized bitemporal latent, and we expect their semantic representations to be similar in the unchanged areas. Therefore, we propose a temporal constraint loss $\mathcal{L}_t$ to measure their temporal similarity, calculated as follows:

$$\mathcal{L}_t(l_1, l_2) = [1 - \text{cosine}(l_1, l_2)] \cdot c \tag{6}$$

where c is the ground-truth (GT) change label on which unchanged areas are annotated as 1. This is to exclude the changed areas from loss calculations. In this way, the task-agnostic semantic representations relevant to the RS domain can be exploited. We further utilize attention operations to embed the semantic focus into change features f_c, before mapping them into a change map m. The change features are obtained by forwarding the SAM features d_1, d_2, andd_3 into a convolutional block. The attention embedding operation is as follows:

$$m = \text{conv}_2\{\sigma[\text{conv}_1(l_1 \oplus l_2)] \cdot f_c\} \tag{7}$$

where $\oplus$ is a channelwise concatenation operation, σ is a sigmoid normalization function, and conv_1 and conv_2 are two convolutional modules to adjust the feature channels. This ensures that the CD results are semantic-aware, thus better segmenting the semantic changes.

D. Implementation Details

In the following, we report the network settings and the training details in the implementation of the SAM-CD.

1) SAM-CD: The FastSAM is constructed using the YOLO-v8 architecture. To obtain higher accuracy, we utilize the default version of FastSAM. Its parameter size is 68 Mb. Since the CD task is quite different from typical segmentation tasks, SAM-CD utilizes only the visual encoder of FastSAM and discards the prompt decoders. In the task-agnostic semantic learning branch, the semantic embedding block is a single 1×1 convolutional layer. The number of semantic channels k is empirically set to 8, considering that there are usually few interesting LCLU classes in typical CD applications. In the CD branch, we leverage six layers of residual convolutional blocks to construct the change embedding module following the practice in [20]. This is to better transform semantic features into change representations.

2) Training Settings: The proposed method is implemented in PyTorch. The training is iterated for 50 epochs. We set the initial learning rate to 0.1 and update it at each iteration as $0.1 \times (1 - \text{iterations}/\text{total_iterations})^{1.5}$. The stochastic gradient descent algorithm is used to optimize the gradient. We apply only simple geometric augmentations to the input images, including random flipping and random cropping. During the inference, we apply a test-time augmentation operation,

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which includes eight times of flipping operations to produce more stable prediction results.

For more details, readers are encouraged to visit our codes released at: https://github.com/ggsDing/SAM-CD.

IV. Experimental Results

This section presents the experimental results obtained on the benchmark datasets. First, we conduct ablation studies to test the effectiveness of the proposed SAM-CD architecture. Then, we test the label-efficient learning capability of the proposed method. Finally, we compare our results with the SOTA methods in CD on VHR RSIs.

A. Experimental Setup

1) Dataset: We conduct experiments on four benchmark CD datasets, including the levir change detection (LEVIR-CD) dataset [37], Wuhan University CD (WHU-CD) dataset [38], China land cover dataset (CLCD), and S2Looking dataset [39]. Below, we briefly introduce these two datasets.

a) LEVIR-CD dataset: This is a large-scale benchmark dataset for CD in VHR RSIs. It consists of Google Earth images collected in six cities in TX, USA, between 2012 and 2016. Most of the changes in this dataset are related to construction growth. A total of 31 333 changed objects are annotated. The spatial resolution of images is 0.5 m per pixel. There are a total of 637 pairs of images, and each has 1024×1024 pixels. The number of image pairs in the training, validation, and test sets is 445, 64, and 128, respectively.

b) WHU-CD dataset: This is an aerial benchmark dataset constructed for building CD. The RSIs were collected in Christchurch, New Zealand, between 2012 and 2016, covering an urban area of 20.5 km^2. The observed area went through an earthquake in 2011; thus, there is a significant growth in the number of constructed buildings (from 12 796 to 16 077). The spatial resolution of images is 0.2 m per pixel. The original image tiles have $32\,507 \times 15\,354$ pixels. Following [26], we divide the dataset into training, validation, and test sets with 6096, 762, and 762 pairs of image patches (each with a spatial size of 256×256, respectively).

c) CLCD dataset: This dataset is constructed focusing on cropland changes. It consists of 600 pairs of GaoFen-2 satellite images collected in Guangdong, China. The two acquisition dates are in 2017 and 2019. Each image has 512×512 pixels, and the spatial resolution is 0.5–2 m. Multiple types of LCLU changes are annotated, including buildings, roads, lakes, and bare lands. The split of training, validation, and test sets is 320, 120, and 120 image pairs, respectively.

d) S2Looking dataset: This is a large-scale benchmark dataset for building CD. It is constructed using VHRs collected by various satellites, including GaoFen, SuperView, and BeiJing-2, covering various rural areas around the world. There are a total of 5000 pairs of images, each has 1024×1024 pixels. The spatial resolution of images is 0.5–0.8 m. Different from other CD datasets, many image pairs in the S2Looking dataset are captured at different imaging angles. This causes spatial misalignment and, thus, poses challenges to CD.

2) Evaluation Metrics: Following literature studies [26], we adopt precision (Pre), recall (Rec), F_1, intersection over union (IoU), and overall accuracy (OA) as the evaluation metrics to assess the accuracy of CD methods. The related calculations are

$$\text{Pre} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{8}$$

$$F_1 = 2 \times \frac{P \times R}{P+R}, \quad \text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{10}$$

where TP, FP, TN, and FN represent true positive, false positive, true negative, and false negative, respectively. It is worth noting that the quantitative results in this section are obtained under the same evaluation settings (including the implementation of calculations) as those in [26].

B. Ablation Study

To quantitatively evaluate the improvements brought by the proposed techniques, we compare the obtained accuracy with the baseline method. It is a network constructed following the common settings in CD, i.e., using two Siamese CNNs to extract temporal features and a decoder to embed the change representations as in [8]. We construct the baseline CNN with two Siamese ResNet34 encoders, since their parameter size (81 Mb) is close to the FastSAM (68 Mb).

The results are presented in Table I. One can observe that compared with the baseline method, adding a plain FastSAM brings significant accuracy improvements. The OA, mean IoU (mIoU), and F_1 increase by 0.23%, 1.83%, and 1.11%, respectively. We also present some examples of the CD results in Fig. 3. One can observe that the proposed techniques in the SAM-CD bring progressive improvements to the segmentation results. Specifically, they boost the recognition of nonsalient changes, e.g., the dark buildings in Fig. 3(a) and (c), and the emerged objects that can be confused with the environment, e.g., the constructions in Fig. 3(b).

1) SAM and FastSAM: In the SAM-CD architecture, the FastSAM encoder can be replaced by other VFMs. Since SAM is the first and the most well-known VFM in the segmentation task, we also tested incorporating SAM encoder in the SAM-CD architecture. Table II reports the performance of the SAM-CD equipped with different versions of the SAM and the FastSAM. Since the SAM is built on ViT and requires a fixed spatial input size of 1024×1024 (smaller images will be upsampled before processing), it is much more expensive in computational resources compared with the FastSAM (built on CNNs). The costs of computer resources are measured by the number of parameters (param) and the floating point of operations (FLOPs). The FLOPs are tested using a pair of input images with a size of $3 \times 256 \times 2$. The computational costs of the SAM-CD with SAM encoders are more than 100 times higher than those of the SAM-CD with FastSAM encoders in FLOPs.

Due to the limitation of computational resources, we only tested the accuracy of the SAM-b and the SAM-l encoders. One can observe that the SAM-CD (with FastSAM encoders)

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TABLE I
QUANTITATIVE RESULTS OF THE ABLATION STUDY (TESTED ON THE LEVIR-CD DATASET)

Methods	Proposed Techniques		Accuracy		
	SAM	$\mathcal{L}_t$	$OA(\%)$	$mIoU(\%)$	$F_1(\%)$
siamese CNN (baseline)			98.82	88.96	93.87
SAM-CD (w/o. $\mathcal{L}_t$)	$\surd$		99.05	90.79	94.98
SAM-CD	$\surd$	$\surd$	99.14	91.68	95.50

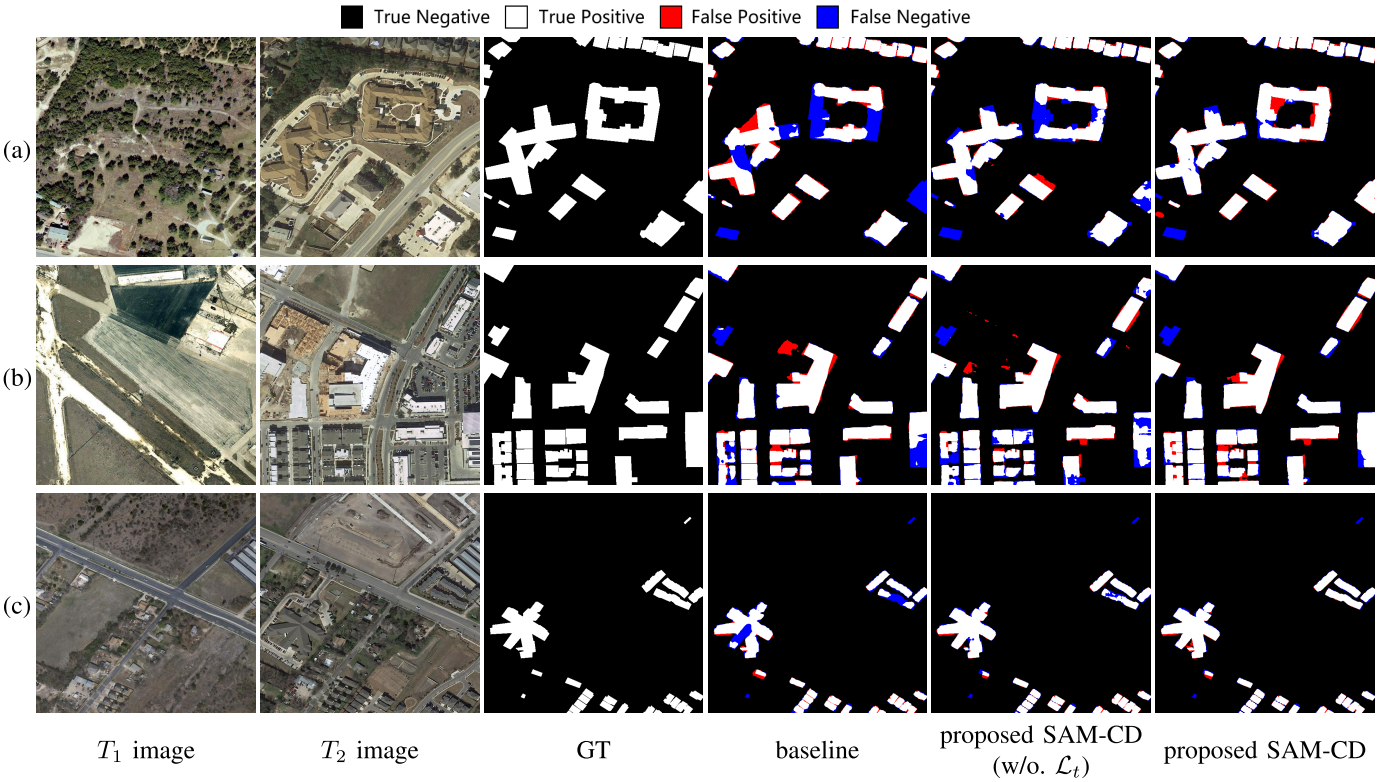


Fig. 3. CD results of the different methods in the ablation study. The predicted maps are compared with the GT maps. The differences are highlighted in color. (a)–(c) Results obtained on three bi-temporal image pairs in the LEVIR-CD dataset.

TABLE II
PERFORMANCE OF THE SAM-CD USING DIFFERENT VISUAL ENCODERS (TESTED ON THE LEVIR-CD DATASET)

Method	Computation Costs		Accuracy		
	Param(Mb)	FLOPs(Gbps)	OA	$F_1(\%)$	$mIoU(\%)$
SAM-CD (SAM-b)	95.32	807.25	96.80	62.29	53.74
SAM-CD (SAM-l)	330.97	2854.39	98.76	77.31	68.24
SAM-CD (SAM-h)	685.90	5941.31	–	–	–
SAM-CD (FastSAM-s)	13.67	8.50	98.97	94.76	90.42
SAM-CD (FastSAM-x)	70.59	8.60	**99.14**	**95.50**	**91.68**

also exhibits better accuracy compared with their counterparts using the SAM encoders. Although the SAM is known to have better capability to exploit image semantics, the FastSAM can provide access to its low-level spatial features, which is crucial for the CD task. Since the FastSAM-x (default encoder in the FastSAM) leads to the highest accuracy, we adopt it as the default visual encoder in the SAM-CD. In pursuit of higher accuracy, readers are also encouraged to apply other VFMs as $\widetilde{U}(\cdot)$ in the SAM-CD architecture.

2) Selection of T: In (5), T in the softmax function is an important parameter to control the diversity of semantic representations. We conduct experiments to select the optimal T and report the results in Table III. One can observe that the increase of T results in variations in the accuracy metrics. The optimal accuracy is obtained with $T = 3$. This is in line with our intuition that a relatively large T results in more softened latent distributions and, thus, better presents the semantics. Therefore, we set $T = 3$ as the default value in the SAM-CD.

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TABLE III
ACCURACY OF SAM-CD OBTAINED WITH DIFFERENT T VALUES (TESTED ON THE LEVIR-CD DATASET)

SAM-CD T values	Accuracy		
	$OA(\%)$	$F_1(\%)$	$mIoU(\%)$
$T=1$	99.10	95.28	91.30
$T=2$	99.11	95.38	91.47
$T=3$	**99.14**	**95.50**	**91.68**
$T=4$	99.09	95.26	91.26
$T=5$	99.12	95.37	91.45

TABLE IV
ACCURACY OF SAM-CD OBTAINED USING DIFFERENT LAYERS OF THE FASTSAM FEATURES (TESTED ON LEVIR-CD)

SAM-CD feature layers	Accuracy		
	$OA(\%)$	$F_1(\%)$	$mIoU(\%)$
$l_1(scale=1/4)$	99.02	94.82	90.53
$l_2(scale=1/8)$	97.10	84.17	75.34
$l_3(scale=1/16)$	98.34	91.15	84.73
$l_4(scale=1/32)$	99.02	94.83	90.54
l_1, l_2, l_3	99.12	95.44	91.57
l_2, l_3, l_4	98.84	93.92	89.05
l_1, l_2, l_3, l_4	**99.14**	**95.50**	**91.68**

3) Selection of Feature Layers: In the SAM-CD architecture, the FastSAM encoder extracts four layers of features corresponding to 1/4, 1/8, 1/16, and 1/32 of the input spatial size. Denote these features as l_1, l_2, l_3, and l_4, respectively. We further experiment to quantitatively evaluate the contribution of each layer of feature and to decide the optimal group of features to use in CD. This is conducted by slightly changing the SAM-CD architecture to use only one layer or a group of layers. The results are reported in Table IV. One can observe that performing CD using l_1 and l_4 leads to a relatively high accuracy. This reveals that both semantic features and spatial features are crucial to CD. In addition, as indicated in the results using l_2, l_3, and l_4, missing the low-level features (l_1) leads to a significant decrease in accuracy. Since the fusion of all four layers of features leads to the highest accuracy, this combination is selected in the SAM-CD architecture.

4) Visualization of the Latent: To intuitively evaluate the effect of the proposed task-agnostic semantic learning, we visualize the semantic latent l in (7) and present it in Fig. 4. One can observe that the learned latent corresponds to various types of LCLU types that are not provided with semantic labels. Apart from buildings that are annotated in the CD datasets, semantic objects, such as ponds, roads, and low vegetation, are attended to in the response maps. This demonstrates that the proposed method can capture the underlying semantics in RSIs, even without providing the corresponding human annotations.

C. Comparative Experiments With Fully Supervised CD Methods

1) Quantitative Results: We compare the accuracy of the proposed SAM-CD with the SOTA CD methods and report the results obtained in four benchmark datasets in Tables V–VII. The compared methods include classic CNN-based methods, such as FC-Siam-diff, FC-Siam-conc [8], and Siamese nested U-shaped network (SNUNet) [40], and recent models based on transformer or self-attention, such as BIT [25], ChangeFormer [41], CTD-Former [26], change guiding network (CGNet) [42], and EATDer [24].

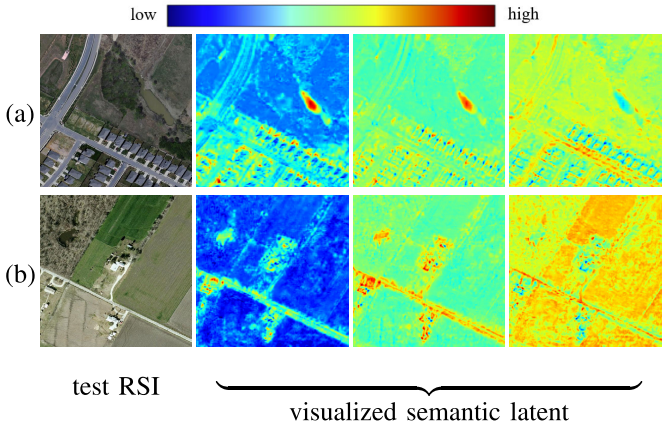


Fig. 4. Visualization of the semantic latent. Warm colors indicate high values and vice versa for cold colors. (a) and (b) Visualization results obtained on two test RSIs in the LEVIR-CD dataset.

Among the literature methods, the recent transformer-based methods (including the BIT, ChangeFormer, CTD-Former, and EATDer) generally obtain higher accuracy. The CTD-Former obtains the third-best mIoU on the WHU-CD and CLCD datasets, while the EATDer and ChangeFormer obtain the third-best accuracy on the LEVIR-CD and S2Looking datasets, respectively. CGNet outperforms most of the literature methods in all four datasets. It also obtains the highest Pre and the highest Rec on the WHU-CD and S2Looking datasets, respectively. However, a common problem of these literature methods is that they are mostly designed for detecting building changes; thus, their capability of modeling LCLU changes and complex change scenes with spatial shift is limited. This can be verified in their relatively lower accuracy in the CLCD and S2Looking datasets.

The proposed SAM-CD achieves robust accuracy improvements over the SOTA methods in all four datasets. In particular, in the results obtained on the CLCD dataset, it surpass the SOTA accuracy by 1.31% and 1.7% in mF_1 and mIoU, respectively. This demonstrates that the semantic learning in SAM-CD, driven by the VFMs, helps to better discriminate the LCLU changes in RS scenes. In the WHU-CD and S2Looking datasets, it also outperforms the compared literature methods by a large margin. It obtains the best Rec in the LEVIR-CD, WHU-CD, and CLCD datasets, which indicates its capability to detect nonsalient semantic changes.

2) Qualitative Results: In Fig. 5, we present some examples of the CD results obtained on the four benchmark datasets. Compared with two recent methods CGNet and EATDer, the proposed SAM-CD exhibits advantages in detecting nonsalient changes, such as new buildings [Fig. 5(a) and (d)], removed house [Fig. 5(c)], new pond [Fig. 5(e)], and constructions [Fig. 5(f)]. It also has fewer false alarms while dealing with complex changes in Fig. 5(b), (e), and (h). This demonstrates that the SAM-CD can better discriminate between temporal differences and semantic changes.

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TABLE V
QUANTITATIVE RESULTS OF DIFFERENT CD METHODS OBTAINED ON THE LEVIR-CD AND WHU-CD DATASETS (%)

Method	Levir-CD					WHU-CD				
	Pre	Rec	OA	mF_1	$mIoU$	Pre	Rec	OA	mF_1	$mIoU$
FC-Siam-diff [8]	92.93	87.02	98.15	89.73	82.67	94.52	89.32	98.70	91.74	85.65
FC-Siam-conc [8]	92.03	89.82	98.28	90.89	84.34	93.94	92.83	98.90	93.37	88.19
SNUNet [40]	93.83	90.11	98.50	91.88	85.84	92.61	83.26	98.10	87.30	79.40
BIT [25]	90.27	83.37	97.60	86.46	78.23	87.92	93.41	98.29	90.46	83.72
ChangeFormer [41]	91.46	86.31	97.95	88.69	81.21	96.28	92.95	99.12	94.55	90.08
CTD-Former [26]	93.60	91.85	98.62	92.71	87.11	96.74	97.03	99.50	96.89	94.11
CGNet [42]	95.95	94.95	99.13	95.44	91.58	**98.04**	96.14	99.52	97.07	94.44
EATDer [24]	**96.29**	92.70	98.87	94.41	89.84	95.57	93.06	99.01	94.28	89.64
SAM-CD (proposed)	95.87	**95.14**	**99.14**	**95.50**	**91.68**	97.97	**97.20**	**99.60**	**97.58**	**95.36**

TABLE VI
QUANTITATIVE RESULTS OF DIFFERENT CD METHODS OBTAINED ON THE CLCD DATASET (%)

Method	Accuracy Metrics				
	Pre	Rec	OA	mF_1	$mIoU$
FC-Siam-diff [8]	83.13	74.60	94.73	78.10	68.19
FC-Siam-conc [8]	81.65	79.94	94.84	80.77	70.98
SNUNet [40]	85.92	82.62	95.84	84.19	75.11
BIT [25]	81.42	70.27	94.17	74.39	64.51
ChangeFormer [41]	84.41	81.36	95.47	82.80	73.40
CTD-Former [26]	87.29	83.17	96.12	85.08	76.24
CGNet [42]	85.87	85.30	96.00	85.58	76.83
EATDer [24]	87.16	77.18	93.84	81.16	71.18
SAM-CD (proposed)	**88.25**	**85.65**	**96.26**	**86.89**	**78.53**

TABLE VII
QUANTITATIVE RESULTS OF DIFFERENT CD METHODS OBTAINED ON THE S2LOOKING DATASET (%)

Method	Accuracy Metrics			
	Pre	Rec	F_1	IoU
FC-Siam-diff [8]	83.49	32.32	46.60	30.38
FC-Siam-conc [8]	68.27	18.52	13.54	-
SNUNet [40]	45.26	50.60	47.78	31.39
BIT [25]	70.26	56.53	62.65	45.62
ChangeFormer [41]	**72.82**	56.13	63.39	-
CGNet [42]	70.18	**59.38**	64.33	47.41
EATDer [24]	65.85	54.74	59.78	42.64
SAM-CD (proposed)	72.80	58.92	**65.13**	**48.29**

3) Computational Costs: Table VIII reports the cost of computational resources of different methods. Among the recent literature methods, the SNUNet has the fewest parameters and the highest inference speed (indicated in infer.). The inference speed of the BIT and CGNet is also relatively high. In the SAM-CD, the FastSAM encoder accounts for most of the parameters (68 m), whereas the proposed network modules account for 2.49-m parameters. The computational efficiency (indicated in FLOPs) of the SAM-CD is much lower than most of the recent methods (including SNUNet, BIT, ChangeFormer, CTD-Former, CGNet, and EATDer). However, the inference speed of the SAM-CD is higher than most of the recent methods except for the ChangeFormer.

TABLE VIII
COMPUTATIONAL COSTS OF DIFFERENT CD METHODS

Method	Param(Mb)	FLOPs(Gps)	Infer.(s)
FC-Siam-diff	1.66	21.45	0.66
FC-Siam-conc	2.74	8.77	0.56
SNUNet	1.35	18.91	0.79
BIT	11.47	19.60	1.12
ChangeFormer	41.03	811.15	10.77
CTD-Former	3.85	104.62	2.62
CGNet	33.68	82.23	1.54
EATDer	6.61	23.46	3.67
SAM-CD	2.49 +68 (FastSAM)	8.60	4.59

D. Comparative Experiments With Semisupervised CD Methods

A valuable feature of CFMs is their generalization to visual tasks. In experiments, we find that the proposed SAM-CD exhibits less dependence on training samples compared with standard deep learning-based methods. Tables IX and X compare the accuracy of different CD methods trained with limited proportions of training data. Apart from the baseline method (ResNet-CD), we also compare with the SOTA methods for semisupervised CD, including semi-supervised semantic segmentation generative adversarial network (S4GAN) [43], SemiCDNet [44], SemiCD [45], and UniMatch [46]. It is worth noting that the reported accuracy of the SAM-CD is obtained using plain supervised training, i.e., without using complex semisupervised training strategies, such as contrastive learning, adversarial learning, and spectralwise augmentations. The reported accuracy in this section is obtained following the experimental setups in [45].

The quantitative results (reported in Tables IX and X) reveal that the SAM-CD outperforms most semisupervised CD methods, except for the very recent method UniMatch [46]. Due to its contrastive learning and feature-level augmentation strategies, the UniMatch obtains the highest accuracy with 5%, 10%, and 20% of the training data. The proposed SAM-CD obtains the highest accuracy while using 40% and 100% of the training data (on the LEVIR-CD dataset) and obtains the second-best accuracy with 5%, 10%, and 20% of the training data. However, there is a drop in accuracy

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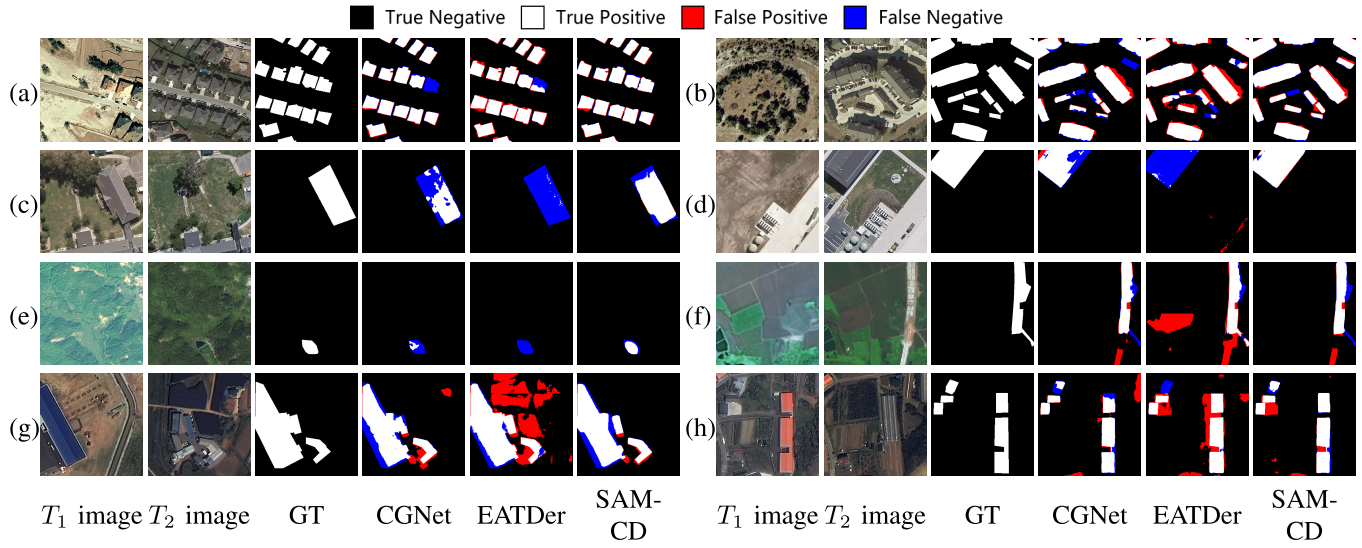


Fig. 5. CD results of the different fully supervised methods. (a) and (b) Results on the LEVIR-CD dataset. (c) and (d) Results on the WHU-CD dataset. (e) and (f) Results on the CLCD dataset. (g) and (h) Results on the S2Looking dataset.

TABLE IX

ACCURACY OF THE SAM-CD AND SEMISUPERVISED CD METHODS VERSUS DIFFERENT PROPORTIONS OF TRAINING DATA (TESTED ON LEVIR-CD). THE EVALUATION METRICS ARE IoU^c AND OA, RESPECTIVELY

Methods	Levir-CD				
	5%	10%	20%	40%	100%
S4GAN [43]	64.0 / 97.89	67.0 / 98.11	73.4 / 98.51	75.4 / 98.62	-
SemiCDNet [44]	67.6 / 98.17	71.5 / 98.42	74.3 / 98.58	75.5 / 98.63	-
SemiCD [45]	72.5 / 98.47	75.5 / 98.63	76.2 / 98.68	77.2 / 98.72	77.9 / 98.77
UniMatch [46]	**80.7 / 98.95**	**82.0 / 99.02**	**81.7 / 99.02**	82.1 / 99.03	-
SAM-CD (proposed)	73.87 / 98.50	79.17 / 98.86	80.18 / 98.91	**83.17 / 99.07**	**84.26 / 99.14**

TABLE X

ACCURACY OF THE SAM-CD AND SEMISUPERVISED CD METHODS VERSUS DIFFERENT PROPORTIONS OF TRAINING DATA (TESTED ON WHU-CD). THE EVALUATION METRICS ARE IoU^c AND OA, RESPECTIVELY

Methods	WHU-CD				
	5%	10%	20%	40%	100%
S4GAN [43]	18.3 / 96.69	62.6 / 98.15	70.8 / 98.60	76.4 / 98.96	-
SemiCDNet [44]	51.7 / 97.71	62.0 / 98.16	66.7 / 98.28	75.9 / 98.93	-
SemiCD [45]	65.8 / 98.37	68.1 / 98.47	74.8 / 98.84	77.2 / 98.96	85.5 / 99.38
UniMatch [46]	**80.2 / 99.15**	**81.7 / 99.22**	**81.7 / 99.18**	**85.1 / 99.35**	-
SAM-CD (proposed)	66.91 / 98.56	71.38 / 98.73	77.92 / 99.00	84.00 / 99.23	**91.15 / 99.60**

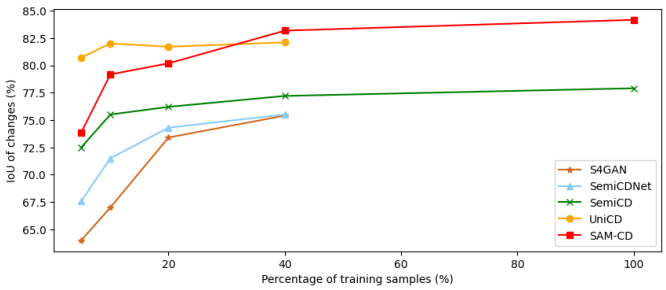


Fig. 6. IoU_c of semisupervised CD methods versus different percentages of training data (tested on the LEVIR-CD dataset).

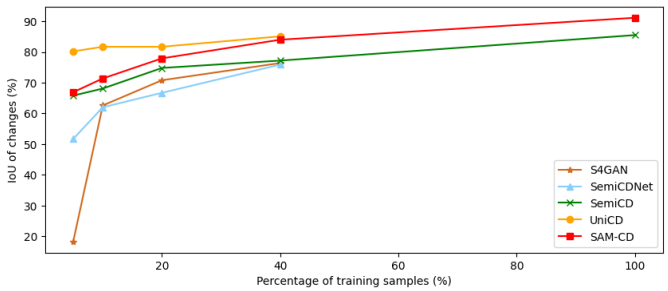


Fig. 7. IoU_c of semisupervised CD methods versus different percentages of training data (tested on the WHU-CD dataset).

while using only 5% of the training samples. We assume that this is because the training of the adaptor and the change head still requires a certain extent of supervised training. We further plot the accuracy versus different training proportions in Figs. 6 and 7. The evaluation metric is the IoU of change areas following the literature studies [46]. It points out that, while a certain number of training samples are still required to adapt the FastSAM to CD task, the

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accuracy of the SAM-CD is comparable to SOTA methods for semisupervised CD.

V. Conclusion

Typical deep learning-based CD methods compare temporal differences to segment the interesting changes; thus, they are often affected by seasonal changes and different imaging conditions. In this article, we propose a SAM-CD architecture that models the semantic latent in VHR RSIs to detect the changed objects. It leverages the FastSAM model to extract visual features in ground objects and utilizes the underlying temporal constraints in RSIs to supervise the learning of task-agnostic semantic representations.

Experimental results tested on four benchmark datasets demonstrate that the proposed SAM-CD gains significant accuracy improvements over the SOTA methods. Due to its capability of modeling semantic latent in VHR RSIs, it can better discriminate between semantic changes and temporal differences. Moreover, benefiting from the great generalization capability of VFM encoders, the SAM-CD exhibits a sample-efficient learning ability. It can obtain fairly accurate results with a limited number of training samples.

However, there are still several remaining problems. First, the proposed SAM-CD still requires fully supervised training. Under the circumstances that there are only very limited training samples, its accuracy is below the SOTA methods in semisupervised CD. Second, the inference speed of the SAM-CD is relatively slow compared with literature methods. To better utilize VFMs in CD tasks, there is a need to reduce the parameters and discard the redundant priors. This can be achieved by using knowledge-distilling learning techniques.

With the advances in VFMs, it is now possible to reach zero-shot or few-shot CD in VHR RSIs with high accuracy. In the future, we will continue to explore combining VFM-based CD with self-supervised training techniques to approach this goal.

References

[1] A. Kirillov et al., "Segment anything," 2023, *arXiv:2304.02643*.

[2] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.

[3] C. Zhang et al., "Faster segment anything: Towards lightweight SAM for mobile applications," 2023, *arXiv:2306.14289*.

[4] W. Ji, J. Li, Q. Bi, T. Liu, W. Li, and L. Cheng, "Segment anything is not always perfect: An investigation of SAM on different real-world applications," 2023, *arXiv:2304.05750*.

[5] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[6] D. Wen et al., "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 68–101, Dec. 2021.

[7] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[8] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[9] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[10] X. Hou, Y. Bai, Y. Li, C. Shang, and Q. Shen, "High-resolution triplet network with dynamic multiscale feature for change detection on satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 103–115, Jul. 2021.

[11] G. Cheng et al., "Change detection methods for remote sensing in the last decade: A comprehensive review," 2023, *arXiv:2305.05813*.

[12] Z. Li, C. Tang, L. Wang, and A. Y. Zomaya, "Remote sensing change detection via temporal feature interaction and guided refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628711.

[13] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, and P. He, "SCDNET: A novel convolutional network for semantic change detection in high resolution optical remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 103, Dec. 2021, Art. no. 102465.

[14] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.

[15] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[16] H. Chen, J. Song, C. Wu, B. Du, and N. Yokoya, "Exchange means change: An unsupervised single-temporal change detection framework based on intra- and inter-image patch exchange," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 87–105, Dec. 2023.

[17] M. Seo, H. Lee, Y. Jeon, and J. Seo, "Self-pair: Synthesizing changes from single source for object change detection in remote sensing imagery," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6363–6372.

[18] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[19] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[20] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[21] J. Wu et al., "A multiscale graph convolutional network for change detection in homogeneous and heterogeneous remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102615.

[22] X. Tang et al., "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5609715.

[23] P. Yuan, Q. Zhao, X. Zhao, X. Wang, X. Long, and Y. Zheng, "A transformer-based Siamese network and an open optical dataset for semantic change detection of remote sensing images," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 1506–1525, Dec. 2022.

[24] J. Ma, J. Duan, X. Tang, X. Zhang, and L. Jiao, "EATDer: Edge-assisted adaptive transformer detector for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602015.

[25] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607514.

[26] K. Zhang, X. Zhao, F. Zhang, L. Ding, J. Sun, and L. Bruzzone, "Relation changes matter: Cross-temporal difference transformer for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611615.

[27] L. Ding, J. Zhang, K. Zhang, H. Guo, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for the semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, early access, 2024.

[28] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[29] W. G. C. Bandara, N. G. Nair, and V. M. Patel, "DDPM-CD: Denoising diffusion probabilistic models as feature extractors for change detection," 2022, *arXiv:2206.11892*.

[30] X. Wang, X. Zhang, Y. Cao, W. Wang, C. Shen, and T. Huang, "SegGPT: Segmenting everything in context," 2023, *arXiv:2304.03284*.

[31] X. Zou et al., "Segment everything everywhere all at once," 2023, *arXiv:2304.06718*.

[32] L. P. Osco et al., "The segment anything model (SAM) for remote sensing applications: From zero to one shot," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 124, Nov. 2023, Art. no. 103540.

[33] J. Wu et al., "Medical SAM adapter: Adapting segment anything model for medical image segmentation," 2023, *arXiv:2304.12620*.

[34] S. Ruder, "An overview of multi-task learning in deep neural networks," 2017, *arXiv:1706.05098*.

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[35] L. Ding and L. Bruzzone, "DiResNet: Direction-aware residual network for road extraction in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10243–10254, Dec. 2021.
[36] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017.
[37] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[38] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.
[39] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, Dec. 2021.
[40] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.
[41] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.
[42] C. Han, C. Wu, H. Guo, M. Hu, J. Li, and H. Chen, "Change guiding network: Incorporating change prior to guide change detection in remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8395–8407, 2023.
[43] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high-and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2019.
[44] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.
[45] W. Gedara Chaminda Bandara and V. M. Patel, "Revisiting consistency regularization for semi-supervised change detection in remote sensing images," 2022, *arXiv:2204.08454*.
[46] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.



Lei Ding received the M.S. degree in photogrammetry and remote sensing from PLA Strategic Support Force Information Engineering University, Zhengzhou, China, in 2016, and the Ph.D. degree (cum laude) in communication and information technologies from the University of Trento, Trento, Italy, in 2021.

He is currently a Lecturer at the PLA Strategic Support Force Information Engineering University. Since 2024, he has been a Post-Doctoral Fellow at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests are related to the recognition and localization of remote sensing data.



Kun Zhu received the B.S. degree in remote sensing science and technology from Henan Polytechnic University, Jiaozuo, China, in 2015, and the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2022.

He is currently a Lecturer with the Institute of Geospatial Information, PLA Strategic Support Force Information Engineering University, Zhengzhou, China. His research interests include object detection, object tracking, and remote sensing image processing.



Daifeng Peng received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

He is currently an Associate Professor with the School of Remote Sensing and Geomatics Engineering, Nanjing University of Information Science and Technology, Nanjing, China. His research interests include machine learning, deep learning, multitemporal image analysis, intelligent interpretation, and information extraction from high-resolution remote-sensing imagery.



Hao Tang received the master's degree from the School of Electronics and Computer Engineering, Peking University, Beijing, China, in 2016, and the Ph.D. degree from the Multimedia and Human Understanding Group, University of Trento, Trento, Italy, in 2021.

He was a Visiting Scholar with the Department of Engineering Science, University of Oxford, Oxford, U.K. He is currently a Post-Doctoral Researcher with the Computer Vision Laboratory, ETH Zürich, Zürich, Switzerland. His research interests are deep learning, machine learning, and their applications to computer vision.



Kuiwu Yang received the M.S. degree in communication and information systems from PLA Strategic Support Force Information Engineering University, Zhengzhou, China, in 2005, and the Ph.D. degree in information security from the Beijing University of Posts and Telecommunications, Beijing, China, in 2012.

He is currently an Associate Professor at the PLA Strategic Support Force Information Engineering University. His research interests are related to big data security and artificial intelligence security.



Lorenzo Bruzzone (Fellow, IEEE) received the Laurea (M.S.) degree (summa cum laude) in electronic engineering and the Ph.D. degree in telecommunications from the University of Genoa, Genoa, Italy, in 1993 and 1998, respectively.

He is currently a Full Professor of telecommunications at the University of Trento, Trento, Italy, where he teaches remote sensing, radar, and digital communications. He is the Founder and the Director of the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento. He is the principal investigator of many research projects. Among the others, he is the Principal Investigator of the Radar for Icy Moon Exploration (RIME) instrument in the framework of the JUpiter ICy moons Explorer (JUICE) mission of the European Space Agency. He is the author (or coauthor) of 215 scientific publications in referred international journals (154 in IEEE journals), more than 290 articles in conference proceedings, and 21 book chapters. His articles are highly cited, as proven from the total number of citations (more than 27 000), and the value of the H-index is 78 (source: Google Scholar). His research interests are in the areas of remote sensing, radar, and synthetic aperture radar (SAR); signal processing; machine learning; and pattern recognition. He promotes and supervises research on these topics within the frameworks of many national and international projects.

Dr. Bruzzone has been a member of the Administrative Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) since 2009. He is the Co-Founder of the IEEE International Workshop on the Analysis of Multi-Temporal Remote-Sensing Images (MultiTemp) series and is currently a member of the Permanent Steering Committee of this series of workshops. Since 2003, he has been the Chair of the SPIE Conference on Image and Signal Processing for Remote Sensing. He was invited as a keynote speaker in more than 30 international conferences and workshops. He was a Distinguished Speaker of the IEEE Geoscience and Remote Sensing Society from 2012 to 2016. He has been the Founder of *IEEE Geoscience and Remote Sensing Magazine*, for which he was an Editor-in-Chief from 2013 to 2017. He was a guest co-editor of many special issues of international journals. He is an editor/co-editor of 18 books/conference proceedings and one scientific book. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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